

- 23** A culture of bacteria was allowed to reproduce using nucleotides containing the heavy isotope of nitrogen (^{15}N). After several generations, all of the bacterial DNA molecules contained heavy nitrogen.

DNA was extracted from a sample of the culture, mixed with caesium chloride solution and spun at high speed in a centrifuge. In this process, DNA molecules of different masses separate into bands at different positions in the centrifuge tube. The heavier the DNA molecules, the closer to the bottom of the centrifuge tube that a band forms.

The diagram shows the position of the DNA molecules containing heavy nitrogen in the centrifuge tube.



The culture of bacteria was then allowed to reproduce using nucleotides containing the light isotope of nitrogen (^{14}N).

DNA samples were taken and separated by centrifugation after the bacteria had divided once and again after the bacteria had divided twice.

In which positions would the DNA be found after the cells had divided once and after the cells had divided twice?

	after dividing once	after dividing twice
A	half at K and half at L	quarter at K, quarter at M and half at L
B	half at K and half at M	quarter at K, quarter at M and half at L
C	all at L	half at K and half at L
D	all at M	half at L and half at M